

the total trip extremely long and expensive for the salesman even though each individual step looked good at the time.

The “greediness” refers to the shortsightedness of this locally optimizing decision process. We can understand these greedy approaches to optimization problems with reference to the problem of trying to find the highest point in a complex, hilly terrain, where “high” points are analogous to better, optimal solutions (short distances in the TSP), and “low” points are analogous to worse, suboptimal solutions (long distances in the TSP). A greedy approach to finding the highest point in a hilly terrain would be to always go up, but that might take us to the top of a little foothill instead of the top of the highest mountain. Sometimes it’s better to go down to the bottom of the foothill in order to start the more important ascent of the bigger mountain. Because greedy algorithms search only for local improvements, they will never allow us to go down and can get us stuck on local extrema. This is exactly the problem discussed in Chapter 3.

With that understanding, we’re finally ready to introduce the idea that will enable us to resolve the local optimization problem caused by greedy algorithms. The idea is to give up the naive commitment to always climbing. In the case of the TSP, we may sometimes have to perturb to worse itineraries so that later we can get the best possible itineraries, just as we go down a foothill in order to ultimately go up a mountain. In other words, in order to do better eventually, we have to do worse initially.

Introducing the Temperature Function

To do worse with the intention of eventually doing better is a delicate undertaking. If we’re overzealous in our willingness to do worse, we might go downward at every step and get to a low point instead of a high one. We need to find a way to do worse only a little, only occasionally, and only in the context of learning how to eventually do better.

Imagine again that we’re in a complex, hilly terrain. We start in the late afternoon and know that we have two hours to find the highest point in the whole terrain. Suppose we don’t have a watch to keep track of our time, but we know that the air gradually cools down in the evening, so we decide to use the temperature as a way to gauge approximately how much time we have left to find the highest point.